



Bharat Academix CodeQuest – Team Proposal

Four AI agents designed to empower families, small businesses, seniors, and students across India. Clear problem → focused solution → feasible tech → measurable impact. Concise roadmap and milestones included for hackathon submission.

Problem Statement

Fragmented access

Citizens struggle to find and use government schemes, healthcare reminders, and relevant financial tools in one place.

Low digital literacy

Seniors and many rural users need simple, trustworthy assistants — not complex apps.

Small business pain

Micro and small enterprises lack easy tools for bookkeeping, customer logs, and workforce management.

Student overload

Students need help organizing notes, finding scholarships/internships, and managing finances and careers.



Proposed Solution – Four Specialized Agents



Amma Saarthi

Government Schemes navigator, Recipe AI, Grocery assistant, Smart reminders, Local community support.



Sharmaji Saarthi

Business insights, Customer log, Expense tracking, Employee management, Family wellness dashboards.



Senior Saarthi

SOS & medical reminders, Memory aids, Digital companion, Benefits assistant for seniors.



Student Saarthi

Productivity coach, Notes summarizer, Scholarship & internship finder, Student finance & career navigator.

Technology Stack



Cloud & Backend

Scalable serverless architecture (AWS/GCP/Azure), REST/GraphQL APIs, containerized microservices.



AI & ML

Transformer-based NLP for summaries and chat, retrieval-augmented generation (RAG) against verified datasets.



Frontend & UX

Progressive Web App + lightweight Android client, voice-first interactions in English + major Indian languages.



Security & Compliance

Data encryption, role-based access, local data residency considerations, privacy-first design.

Unique Value & Differentiators

Localized, multi-lingual AI

Knowledge of regional schemes, vernacular language support, and culturally aware responses.

Modular agent architecture

Reusable components (reminders, RAG toolkit, authentication) accelerate development and scaling.

Low-friction experience

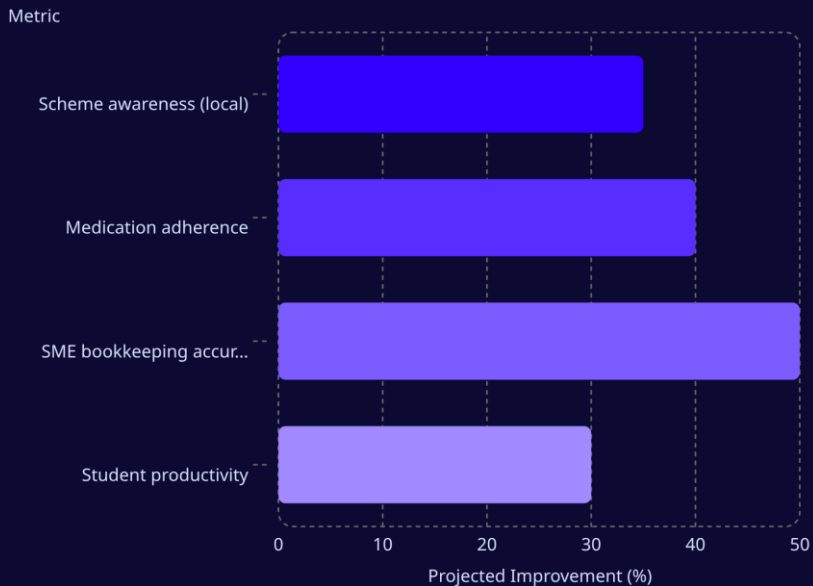
Voice-first, simple flows for seniors and low-literacy users plus optional advanced features for power users.

Impact-focused metrics

Designed to increase scheme uptake, improve SME efficiency, reduce missed medications, and boost student outcomes.



Expected Impact & KPIs



Key performance indicators to measure:

- Active monthly users by segment
- Average time saved per task (minutes)
- Number of successful scheme enrollments assisted
- Reduction in missed medication events

□ All KPIs will be tracked with privacy-preserving analytics and consented telemetry.

Project Roadmap (MVP → Pilot)



Feasibility, Risks & Mitigations

Data quality risk

Mitigation: curate verified government sources, apply RAG with source attribution, human-in-the-loop validation.

Adoption challenges

Mitigation: voice interfaces, local partnerships, pilot feedback loops, simple onboarding flows.

Privacy & compliance

Mitigation: minimal data retention, opt-in telemetry, encryption, and local legal review.

Scalability

Mitigation: microservices, serverless scaling, caching for common queries, and edge-friendly features.





Closing – Why This Matters

Four agents. One unified mission: make essential services, business tools, healthcare reminders, and student support accessible across India. Scalable, respectful of privacy, and built for impact.